% of total fuel
mass
Factor accounting for fuel (expressed in % of total fuel mass consumed in the energy converter) which escapes from the energy converter without being oxidized (including fuel that escapes from combustion chamber/oxidation process and from crankcase, as appropriate)
Cfug % of fuel mass Factor accounting for the fuel (expressed in % of mass of the fuel delivered to the ship) which escapes between the tanks up to the energy converter which is leaked, vented or otherwise lost in the system7
Csfx

gGHG/g fuel Factor accounting for the share of GHG in the components of the fuel (expressed in g GHG/g fuel)
Example: for LNG this value is 1
CfCO2

gCO2/g fuel CO2 emission conversion factor (gCO 2/g fuel completely combusted) for emissions of the combustion and/or oxidation process of the fuel used by the ship
CfCH4

gCH4/g fuel CH4 emission conversion factor (gCH 4/g fuel delivered to

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SOURCE: MEPC.376(80).pdf
gCO2/g fuel CO2 emission conversion factor (gCO 2/g fuel completely combusted) for emissions of the combustion and/or oxidation process of the fuel used by the ship
CfCH4

gCH4/g fuel CH4 emission conversion factor (gCH 4/g fuel delivered to the ship) for emissions of the combustion and/or oxidation process of the fuel used by the ship8
CfN2O

gN2O/g fuel N2O emission conversion factor (gN 2O/g fuel delivered to the ship) for emissions of the combustion and/or oxidation process of the fuel used by the ship
GWPCH4

gCO2eq/g CH4 Global Warming Potential of CH4 over 100 years (based on the fifth IPCC Assessment Report 5)9
Definition as per https://www.ipcc.ch/assessment-report/ar5/

7 Pending further methodological guidance to be developed by the Organization to determine appropriate factor(s), the value of Cfug should be set to zero.

8 For LNG/CNG fuel, the Cslip_engine is covering the role of CfCH4, so CfCH4 is set to zero for these fuels.

9 Set at 28 based on IPCC AR5.

RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
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SOURCE: MEPC.376(80).pdf
9 Set at 28 based on IPCC AR5.

RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
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Term Units Explanation
GWPN2O

gCO2eq/g N2O Global Warming Potential of N2O over 100 years (based on the fifth IPCC Assessment Report 5) .10 Definition as per https://www.ipcc.ch/assessment-report/ar5/
gCO2eq/g GHG Global Warming Potential of GHG in the components of the fuel over 100 years (based on the fifth IPCC scientific Assessment Report)

0 or 1 Carbon source factor to determine whether the emissions credits generated by biomass growth are accounted for in the calculation of the TtW value

gCO2eq/g fuel Emissions credits generated by biomass growth
11

gCO2eq/g fuel Emission credits from the used captured CO 2 as carbon stock to produce synthetic fuels in the fuel production process and utilization (that was not accounted under e fecu and ep)
12

0 or 1 Carbon source factor to determine whether the emissions credits from the used captured CO 2 as carbon stock to

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SOURCE: MEPC.376(80).pdf
process and utilization (that was not accounted under e fecu and ep)
12

0 or 1 Carbon source factor to determine whether the emissions credits from the used captured CO 2 as carbon stock to

produce synthetic fuels in the fuel production process are accounted for in the calculation of the TtW value

13
gCO2eq / g fuel Emission credit from carbon capture and storage (e occs), where capture of CO2 occurs onboard. This should properly account for the emissions avoided through the capture and sequestration of emitted CO 2, if CCS occurs on board.
From the above -mentioned emission credit, all the emissions resulting from the process of capturing (ecc), and transporting (et) the CO 2 up to the final storage (including the emissions related to the injection, etc.) need to be deducted.
This element should be calculated with the following formula:

=

gCO2 / g fuel Credit equivalent to the CO 2 captured and stored (long -term: 100 years)
gCO2eq / g fuel Any emission associated with the process of capturing, compress and temporarily store on board the CO2

10 Set at 265 based on IPCC AR5.

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SOURCE: MEPC.376(80).pdf
term: 100 years)
gCO2eq / g fuel Any emission associated with the process of capturing, compress and temporarily store on board the CO2

10 Set at 265 based on IPCC AR5.

11 Pending further methodological guidance to be developed by the Organization, the value of the multiplication × should be set to zero.

12 Pending further methodological guidance to be developed by the Organization, the value of the multiplication × should be set to zero.

13 Pending further methodological guidance to be developed by the Organization, the value of should be set to zero.

Term Units Explanation
gCO2eq / g fuel Emissions associated with transport to long -term storage site
gCO2eq / g fuel Any emission associated with the process of storing (long-term: 100 years) the captured CO 2 (including fugitive emissions that may happen during long -term storage

and/or the injection of CO2 into the storage)

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SOURCE: MEPC.376(80).pdf
site
gCO2eq / g fuel Any emission associated with the process of storing (long-term: 100 years) the captured CO 2 (including fugitive emissions that may happen during long -term storage and/or the injection of CO2 into the storage)
gCO2eq / g fuel Any additional emission related to the CCS
LCV MJ/g Lower Calorific Value is the amount of heat that would be released by the complete combustion of a specified fuel

5.3 In order to have LCA guidelines that will allow for their clear, robust and consistent application to any possible measure, the methodology allows to calculate two TtW values as follows:

.1 TtW GHG intensity value 1: calculated regardless of the carbon source, therefore the e c and eccu parameters should not be taken into account and the SFc and SFccu value should be always 0; and

.2 TtW GHG intensity value 2: calculated taking into account the carbon source